Dataset Title

This dataset contains a collection of MATLAB and Python scripts for data visualization and analysis. The package includes several directories with MATLAB code and a Python package with a setup.py script that creates executable commands for running specific visualizations as detailed below.

Visualization scripts have been tested in MATLAB2023b and python 3.9 and 3.12

Directory Contents

- **figureX**: MATLAB and Python scripts for visualization routines of main figure results.
- **supp_figureX**: Additional visualization scripts for supplememntray figure results.
- **Shared_data**: Data used across multiple scripts.
- **Utilities**: Helper functions used across the dataset.

Using the MATLAB Code

To run the MATLAB scripts included in this dataset, it is recommended that you add the entire dataset (including all subdirectories) to your MATLAB path. You can do this using the `genpath` function. For example, if your dataset is located in `C:\path\to\dataset`, open MATLAB and run:

```
% Add the entire dataset (and all its subdirectories) to the MATLAB path:
addpath(genpath('C:\path\to\dataset'));

% Optionally, save the updated path for future sessions:
savepath;
```

These commands ensures that MATLAB can locate all functions and scripts distributed across the various directories. Once the paths are set as suggested, any function across the directories can be run either from your editor window or MATLAB command line.

Using the Python Package

The Python portion of the dataset includes a `setup.py` script that defines the package and creates executables (via entry points) for some of the visualization routines.

Using the below instructions assumes you have a python installation and working pip installation utility on your machine.

Installation

- Open a Terminal/Command Prompt
- Navigate to the directory containing `setup.py`:

```
cd /path/to/dataset
  pip install .
```

Running the Executables

After installation, the entry points specified in `setup.py` will be available as command-line executables. The `setup.py` has defined entry points including:

- `run_figure3_F_G_code`
- `run_figure4_E_code`
- `run_supp_figure10_A_code`

you can run them by simply typing the following in your terminal:

```
run_figure3_F_G_code
```

or

```
run_figure4_E_code
```

or

```
run_supp_figure10_A_code`
```

with the last executable including an optional flag '-m' or '--matched' to toggle the matched and unmatched distribution visualizations. Each command will execute the main function in the respective module, launching the corresponding visualization routine.